

Chapter 9

Relationships Between Industries: The Forces Moving Us Toward Long-Run Equilibrium

Learning Objectives

- 9.1 Describe the rate of returns for firms in a competitive industry.
- 9.2 Identify the mean reversion exhibited by profits.
- 9.3 Differentiate between compensating wage differentials and compensating risk differentials.
- 9.4 Explain the investor's decision to move out of risky assets when the risk premia become too small.
- 9.5 Summarize the impact of entry and imitation on eroding profits for monopoly firms.
- 9.6 Analyze how monopoly firms determine price and output, identify barriers to entry, and explain how monopoly affects profits, consumer surplus, and market efficiency.

Chapter 9 – Main Points

- A competitive firm can earn positive or negative profit in the short run until entry or exit occurs. In the long run, competitive firms are condemned to earn only an average rate of return.
- Profit exhibits what is called **mean reversion**, or “regression toward the mean.”.
- If an asset is mobile, then in equilibrium the asset will be indifferent about where it is used (i.e., it will make the same profit no matter where it goes). This implies that unattractive jobs will pay compensating wage differentials, and risky investments will pay compensating risk differentials (or a risk premium).
- The difference between stock returns and bond yields includes a compensating risk premium. When risk premia become too small, some investors view this as a time to get out of risky assets because the market may be ignoring risk in pursuit of higher returns.
- **Monopoly** firms can earn positive profit for a longer period of time than competitive firms, but entry and imitation eventually erode their profit as well.

Warm-Up

- Why do some industries stay profitable for years, while others see profits disappear quickly? What's the difference?

Part One

Perfect Competition

Perfect Competition Characteristics

- Price taker
 - Each firm has to take the market price
- Homogenous good
 - Goods are identical
- Many buyers and many sellers
- Free entry and exit
 - No barriers to entry
- Zero long run economic profits (mean reversion)
 - Normal profits/normal rate of return
 - competition forces inefficient firms out (short run losses) or attracts other firms in (short run profits)

The Revenue of a Competitive Firm

- Total revenue (TR)

$$TR = P \times Q$$

- **Average revenue (AR)**

$$AR = \frac{TR}{Q} = P$$

- **Marginal revenue (MR)**: The change in TR from selling one more unit.

$$MR = \frac{\Delta TR}{\Delta Q} = \frac{TR_{new} - TR_{old}}{Q_{new} - Q_{old}}$$

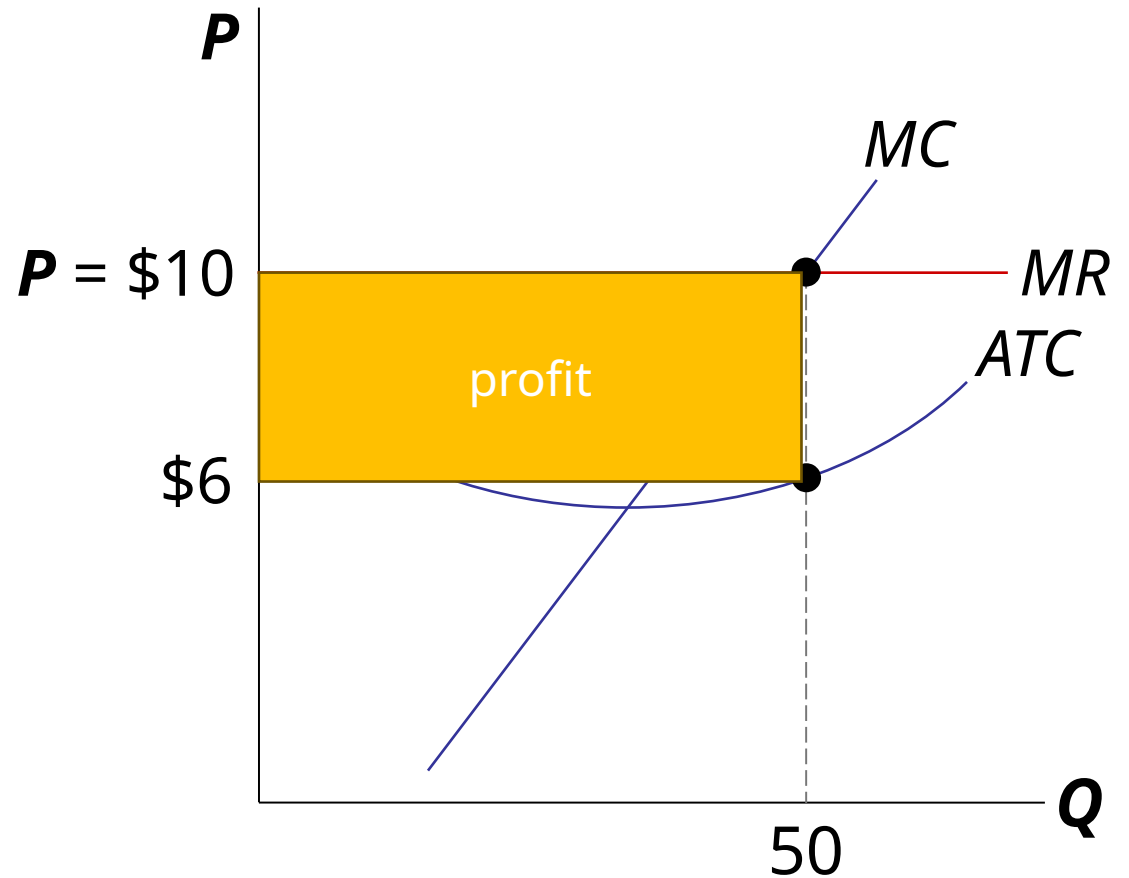
Fill in the empty spaces of the table

Q	P	$TR = P \times Q$	$AR = \frac{TR}{Q}$	$MR = \frac{\Delta TR}{\Delta}$
0	\$10	\$0	n/a	
1	\$10	\$10	\$10	\$10
2	\$10	\$20	\$10	\$10
3	\$10	\$30	\$10	\$10
4	\$10	\$40	\$10	\$10
5	\$10	\$50	\$10	\$10

Identifying a Firm's Profit

Determine this firm's total profit.

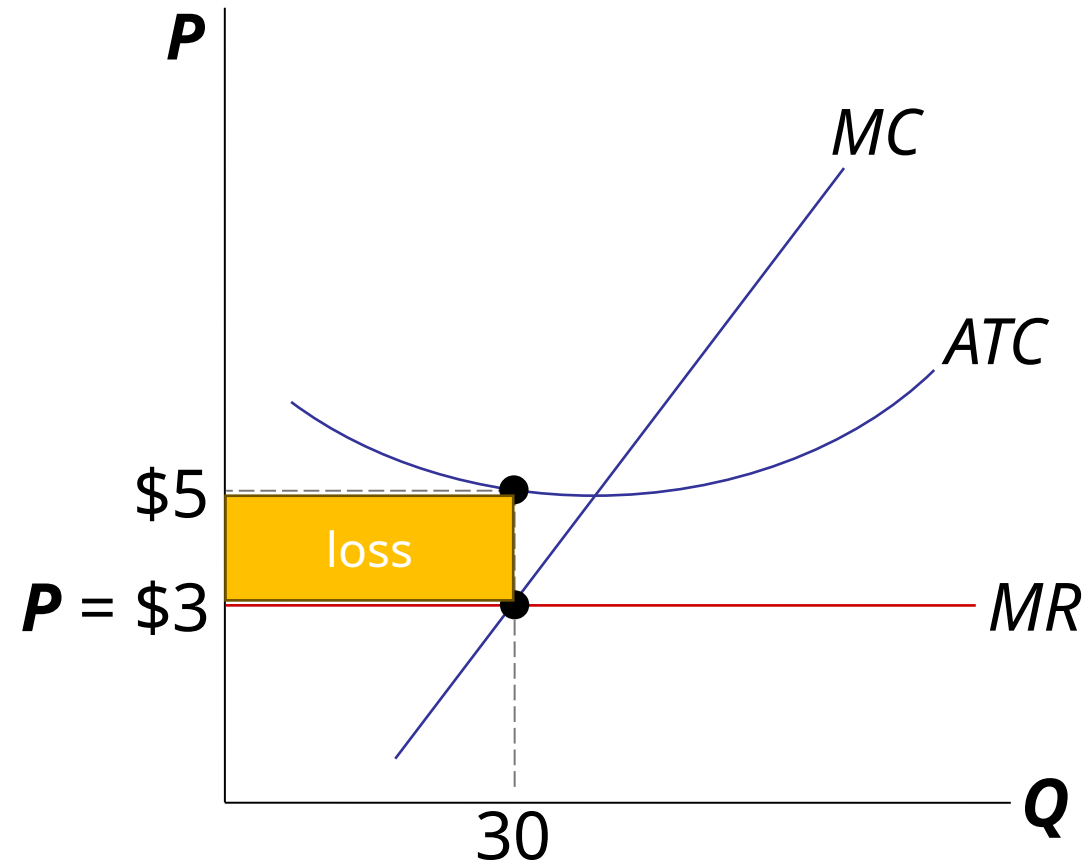
Identify the area on the graph that represents the firm's profit.



Identifying a Firm's Loss

Determine this firm's total loss, assuming $AVC < \$3$.

Identify the area on the graph that represents the firm's loss.



Example (not on a quiz or test)

- A perfectly competitive firm faces a total cost of $\$1000 + 125Q - 0.5Q^2$ and a price of \$100.
- What is the firm's profit maximizing quantity?

$$TC = \$1000 + 125Q - 0.5Q^2$$

$$P = \$100 = MR$$

$$\frac{\partial TC}{\partial Q} = 125 - Q = MC$$

$$125 - Q = 100$$

$$Q = 25$$

$$MR = MC$$

What is the firm's profit? (not on a quiz or test)

$$\pi = TR - TC$$

$$\pi = \$100 * 25 - [1000 + 125(25) - 0.5(25)^2]$$

$$\pi = 2500 - 1000 - 3125 + 312.5$$

$$\pi = -\$1,312.50$$

Example (not on a quiz or test)

- A perfectly competitive firm faces a total cost of $\$1000 + 100Q - 5Q^2 + 1/3Q^3$ and a price of \$300.
- What is the firm's profit maximizing quantity?

$$TC = \$1000 + 100Q - 5Q^2 + 1/3Q^3$$

$$P = \$300 = MR$$

$$MR = MC$$

$$\frac{\partial TC}{\partial Q} = 100 - 10Q + Q^2 = MC$$

$$100 - 10Q + Q^2 = 300$$

$$Q^2 + 10Q - 200 = 0$$

$$(Q + 20)(Q - 10) = 0$$

$$Q = 10$$

What is the firm's profit? (not on a quiz or test)

$$\pi = TR - TC$$

$$\pi = \$300 * 10 - [1000 - 100(10) + 5(10)^2 + \frac{1}{3}(10)^3]$$

$$\pi = 3000 - 1000 + 1000 - 500 - 333.33$$

$$\pi = \$2,166.67$$

Entry and Exit in the Long Run

- In the LR, the number of firms can change due to entry & exit.
- If existing firms earn positive economic profit,
 - new firms enter, SR market supply shifts right.
 - **P** falls, reducing profits and slowing entry.
- If existing firms incur losses,
 - some firms exit, SR market supply shifts left.
 - **P** rises, reducing remaining firms' losses.

The Zero-Profit Condition

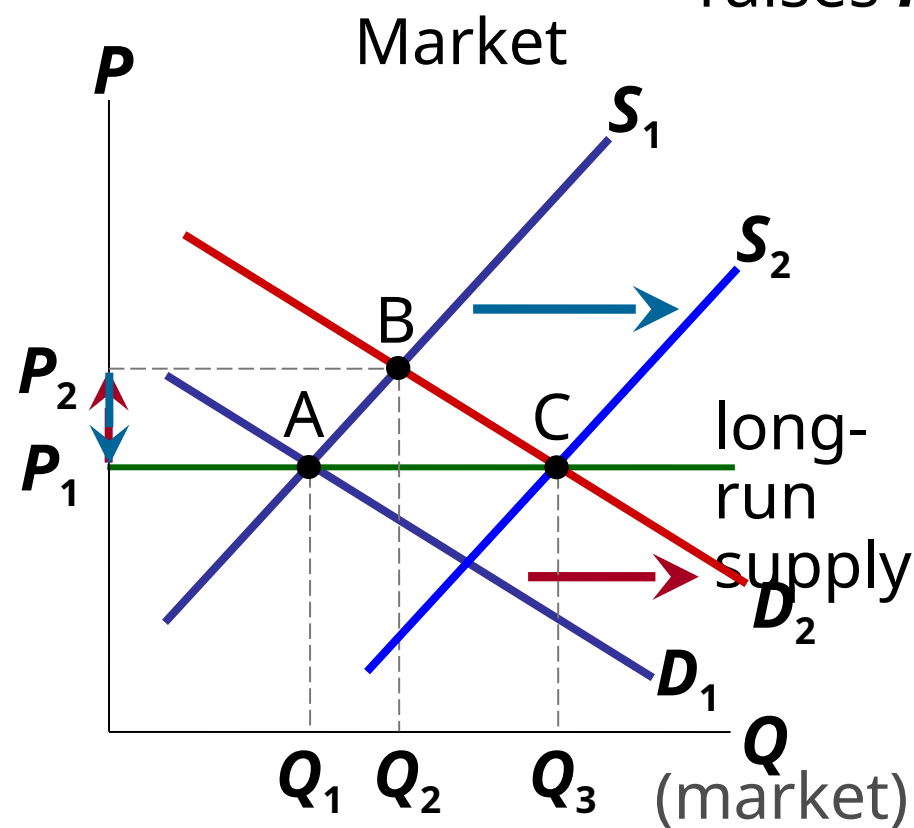
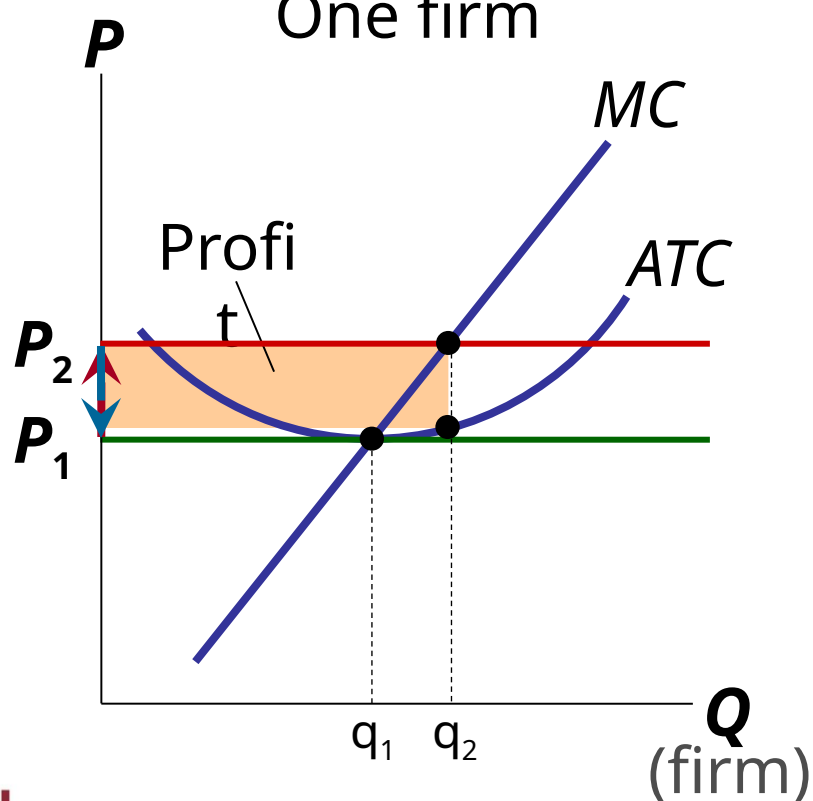
- **Long-run equilibrium:**

The process of entry or exit is complete—remaining firms earn zero economic profit.

- Zero economic profit occurs when $P = ATC$.
- Since firms produce where $P = MR = MC$, the zero-profit condition is $P = MC = ATC$.
- Recall that MC intersects ATC at minimum ATC .
- Hence, in the long run, $P = \text{minimum } ATC$.

SR & LR Effects of an Increase in Demand

A firm ...leading to ...driving profits to zero ...induce entry, but then an
 long-run eq'm. profits for and restoring long-run eq'm. might, reducing increase in demand
 firm. raises P ,...



Mean Reversion

- In competitive industries, profits are magnetic: unusually high profits pull in new rivals, while losses push firms out until only a normal return remains.
- Short-run profits attract new competitors, increasing supply and lowering prices.
- Short-run losses drive firms to exit, reducing supply and increasing prices.
- Over time, economic profits tend to "revert to the mean"—normal levels across industries.
- Resources (labor, capital) reallocate to higher-return uses until equalized returns.

Mean Reversion

Phase	What Happens	Profit Path
1. Positive shock (innovation, demand surge)	Early movers earn economic profit above the opportunity cost of capital.	Profit line spikes above trend.
2. Entry & imitation	New firms copy technology or products. Short-run supply shifts right → price falls.	Profit line slopes downward.
3. Resource reallocation	Labor & capital migrate toward the high-profit sector until marginal return = average return elsewhere.	Profit line crosses the long-run average.
4. Negative shock / loss	Loss-making firms exit; capacity contracts; price rises.	Profit line climbs back to average.

Activity – What Happens to High Profits?

Scenario:

- A new restaurant in a busy city earns a 25% profit margin in Year 1. Word spreads fast, and there are no legal restrictions on opening competitors nearby.

Questions:

1. What do you expect will happen to this restaurant's profits in Year 2 and beyond?
2. What market forces drive this change?
3. How does this relate to mean reversion?
4. What could the restaurant do to maintain above-average profits?

Activity – What Happens to High Profits?

1. Profits are likely to decline as new competitors enter the market, attracted by the high margins.
2. Entry of competitors increases supply, customer choices expand, reducing pricing power, costs may increase (e.g., wages, rent) due to higher demand for local inputs, competitive pressure reduces profit margin
3. Profits in competitive markets tend to revert toward the “normal” rate of return as entry and imitation erode temporary advantages.
4. Build a strong brand or loyal customer base, differentiate through unique offerings or service, create barriers to entry (e.g., exclusive location, supplier contracts), innovate ahead of imitators

Discussion Activity 1

- Why do economic profits in competitive industries tend to disappear over time?
- Think about entry, imitation, and how market forces attract capital to high-return sectors.

Compensating Differentials

- **Compensating wage differential:** Extra pay to attract workers to unpleasant, dangerous, or undesirable jobs.
 - *Example:* Coal miners earn more due to job risk and isolation.
 - Empirical rule of thumb: A 1-percentage-point increase in on-the-job fatality risk raises average wages by $\approx 10\%$
 - **Risk premium:** Higher expected return on risky assets to compensate for uncertainty.
 - *Example:* Stocks offer higher returns than bonds due to greater volatility.
- $$\text{Risk premium} = E[r_{\text{asset}}] - r_{\text{risk-free}}$$
- Since 1928, U.S. equities have delivered ~ 5 percentage points above 3-month T-bills
 - These differentials help equalize overall utility or expected value across options.
 - When market euphoria compresses the risk premium below historical norms, rational investors rotate to bonds, cash, or hedge strategies.
 - *Example:* 2006–07 credit spreads $< 1\%$; many funds de-risked before the 2008 crisis.

Activity – Investor Risk Premium

Scenario:

- In 2021, the average return on U.S. stocks was 8%, while 10-year treasury bonds yielded 1.5%.

Questions:

1. What is the equity risk premium in this case?
2. Would this premium be attractive to investors? Why or why not?
3. Suppose the stock market becomes more volatile but expected return stays at 8%.
 - What happens to investor interest in stocks?
4. What would happen if the premium shrinks to only 2%?

Activity – Investor Risk Premium

1. Equity Risk Premium = $8\% - 1.5\% = 6.5\%$
2. Yes. That's a historically high premium, suggesting strong compensation for risk.
3. If risk rises (e.g., volatility or fear), investors may demand an even higher premium or move to safer assets.
4. A 2% premium might be too small for the risk involved, causing capital to shift to bonds or other low-risk assets.

Discussion Activity 2

- Would you take a more dangerous or unpleasant job for higher pay? Why or why not?
- Consider utility trade-offs — what are you being compensated for?

Part Two

Monopoly

Monopoly Characteristics

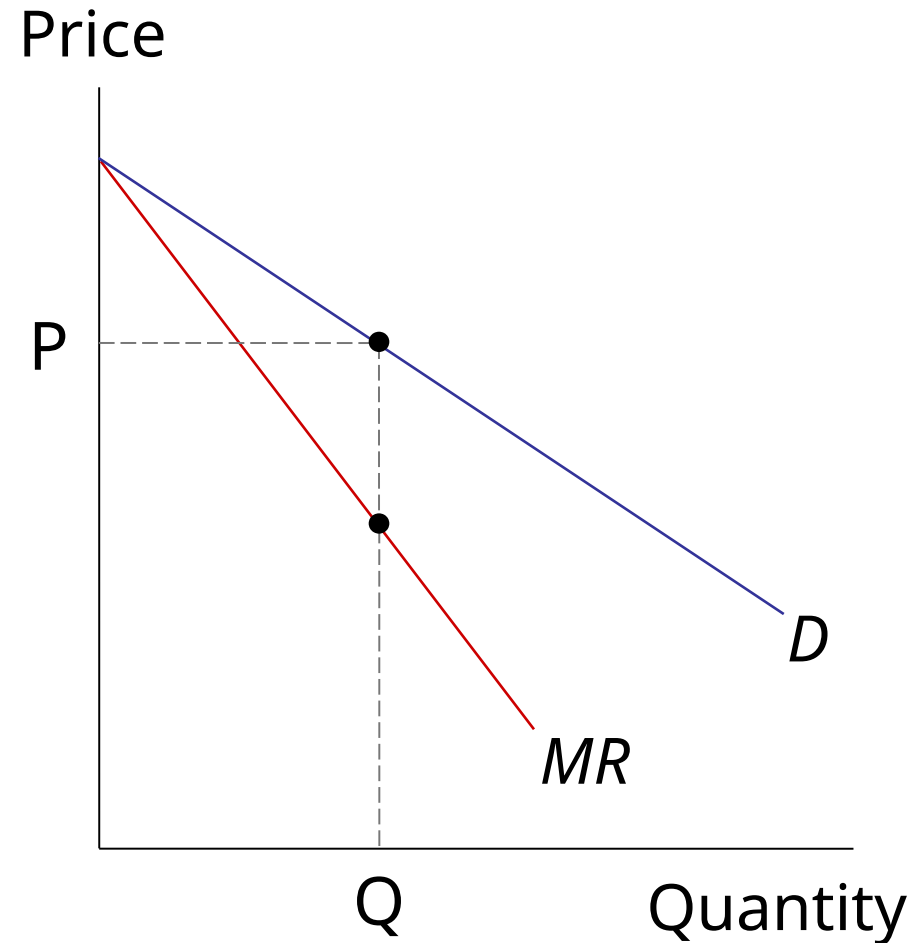
- Price maker/searcher
 - The monopoly makes the price to maximize profit
- One seller
 - No close substitutes
- Restricted entry (barriers to entry)
 - Control of resource, economies of scale, exclusive rights to sell (e.g. patents)
- Long run economic profits
 - May eventually face competition

Why Monopolies Arise

- The main cause of monopolies is **barriers to entry**—other firms cannot enter the market.
- Three sources of barriers to entry:
 - 1. A single firm owns a key resource.
 - *e.g.*, DeBeers owns most of the world's diamond mines, Catholic Church in Medieval Times
 - 2. The govt. gives a single firm the exclusive right to produce the good.
 - *e.g.*, patents, copyright laws
 - 3. **Natural monopoly**: a single firm can produce the entire market Q at lower cost than could several firms.

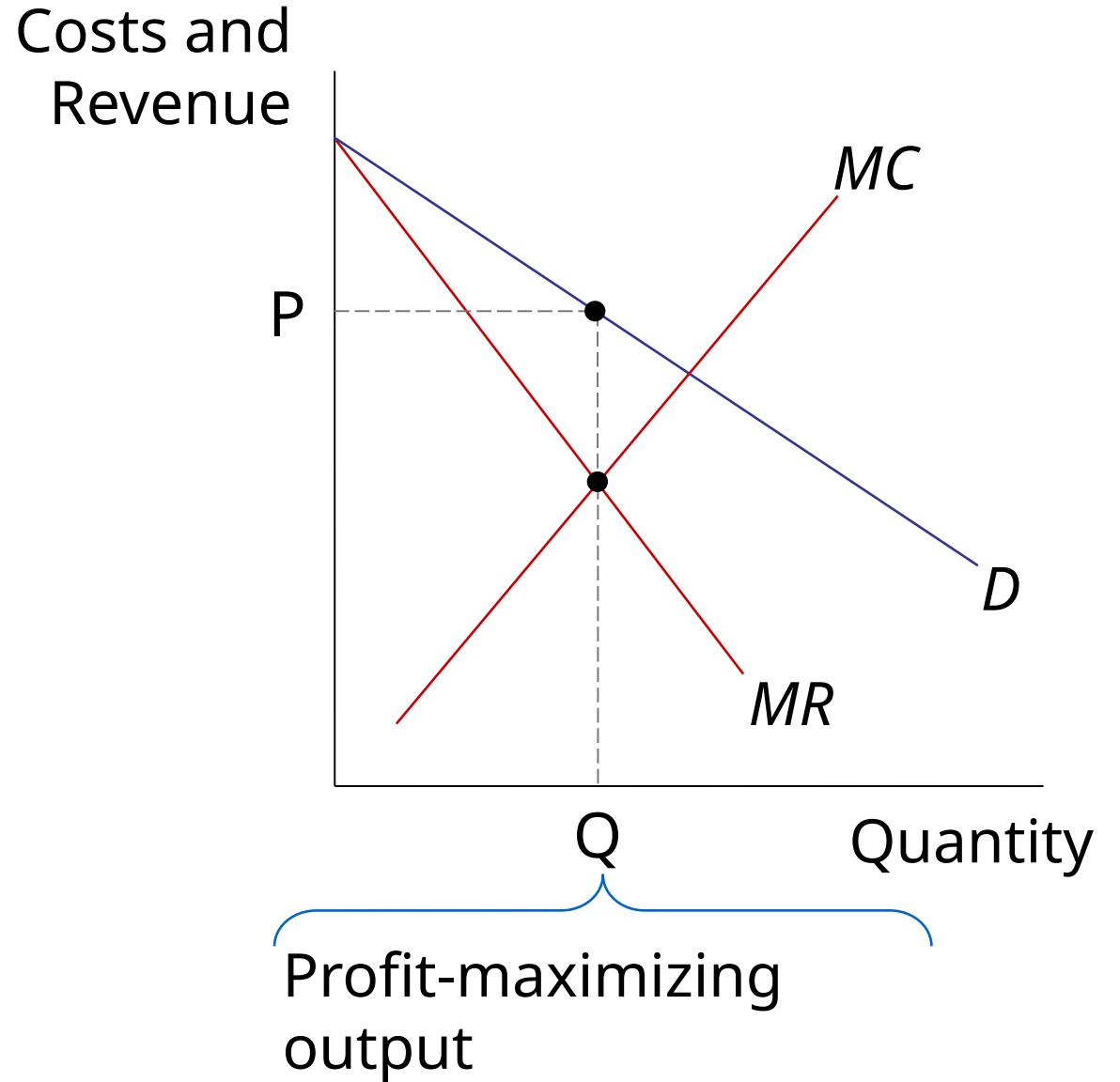
Monopoly Demand

- Recall the monopoly has few close substitutes and is therefore a price maker.
- This causes the monopoly to have a downward sloping demand and marginal revenue curve.



Profit Maximization

- The profit-maximizing Q is where $MR = MC$.
- Find P from the demand curve at this Q .

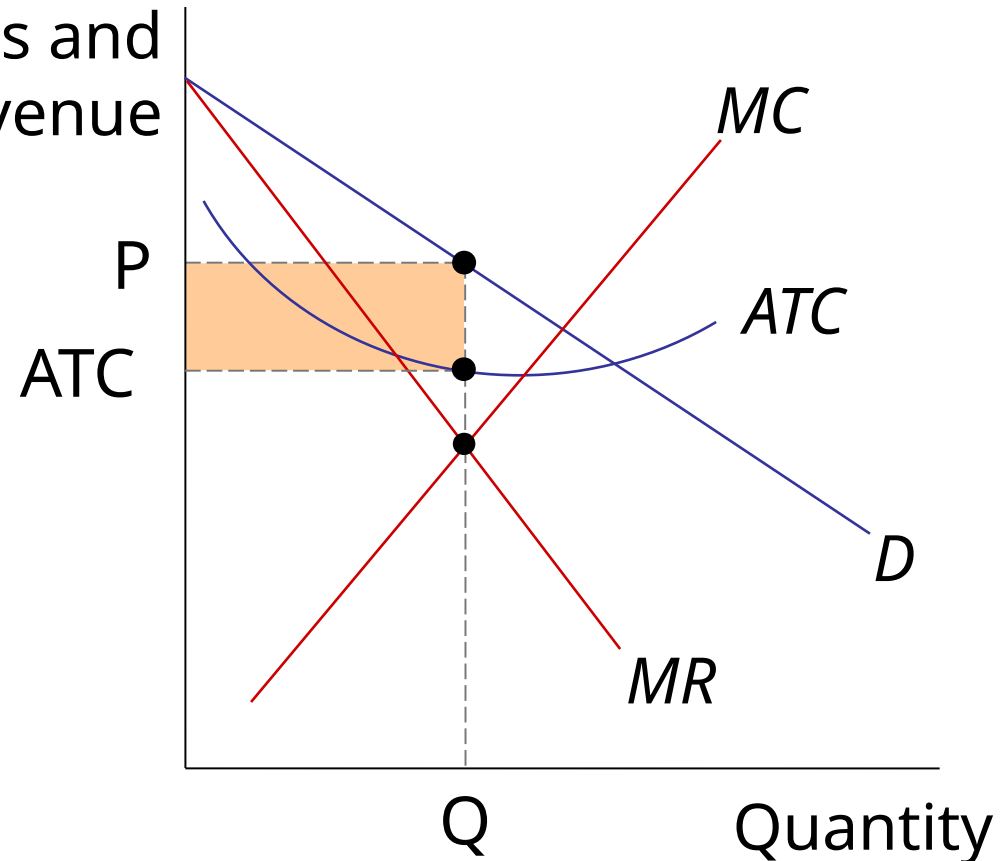


Monopoly Demand and Profit

- As with the competitive firm, the monopolist's profit equals

$$(P - ATC) \times Q$$

- Profits in the long run?
 - Not necessarily.
 - A monopoly can't keep competition out of the market completely. There is too much movement in assets and too much copying of products for this to happen.



Erosion of Profits

Even monopolies can't escape competition forever—innovation, imitation, and market evolution chip away at dominance.

- Temporary Power: Monopolies earn long-run profits only while barriers to entry remain strong (e.g., patents, scale, regulation).
- Profit Erosion Mechanisms:
 - Direct entry – new competitors enter once barriers fall (e.g., patent expires).
 - Imitation – other firms offer similar or substitute products.
 - Substitution – consumers shift to alternative solutions, even if not direct substitutes (e.g., streaming vs. DVDs).
 - Tech leapfrogging – new entrants bypass monopolist's advantages (e.g., ride-sharing bypassed taxi medallions).

Erosion of Profits

Real-World Examples

- Pharmaceuticals: Patent expires → generics enter → price drops 80–90%.
- NYC Taxi Medallions: Monopoly eroded by Uber/Lyft → medallion value collapsed from \$1M+ to <\$100K.
- Blockbuster vs. Netflix: Blockbuster controlled video rentals until streaming changed the game.

Discussion Activity 3

- Why can't monopolies hold on to profits forever? Can you think of examples?
- Think about technology, regulation, or changing consumer behavior (e.g., taxis vs. Uber).

Monopoly Example (not on a quiz or test)

$$Q = \$6,680 - 0.4P$$

$$TC = \$200,000 + 2000Q - 5.5Q^2 + 2Q^3$$

What price and quantity will the monopoly set to maximize profits?

$$P = 16,700 - 2.5Q$$

$$16700 - 5Q = 2000 - 11Q + 6Q^2$$

$$P = 16,700 - 2.5Q$$

$$TR = (16,700 - 2.5Q)Q$$

$$14700 = -6Q + 6Q^2$$

$$P = 16,700 - 2.5(50)$$

$$TR = 16700Q - 2.5Q^2$$

$$-6Q^2 + 6Q + 14700 = 0$$

$$P = 16,575$$

$$\frac{\partial TR}{\partial Q} = 16700 - 5Q$$

$$-Q^2 + Q + 2450 = 0$$

$$\frac{\partial TC}{\partial Q} = 2000 - 11Q + 6Q^2$$

$$(-Q + 50)(Q + 49) = 0$$

$$Q = 50$$

Suppose the monopoly wanted to maximize total revenue. What price should it charge? (not on a quiz or test)

$$\frac{\partial TR}{\partial Q} = 16700 - 5Q$$

$$Q = 3340$$

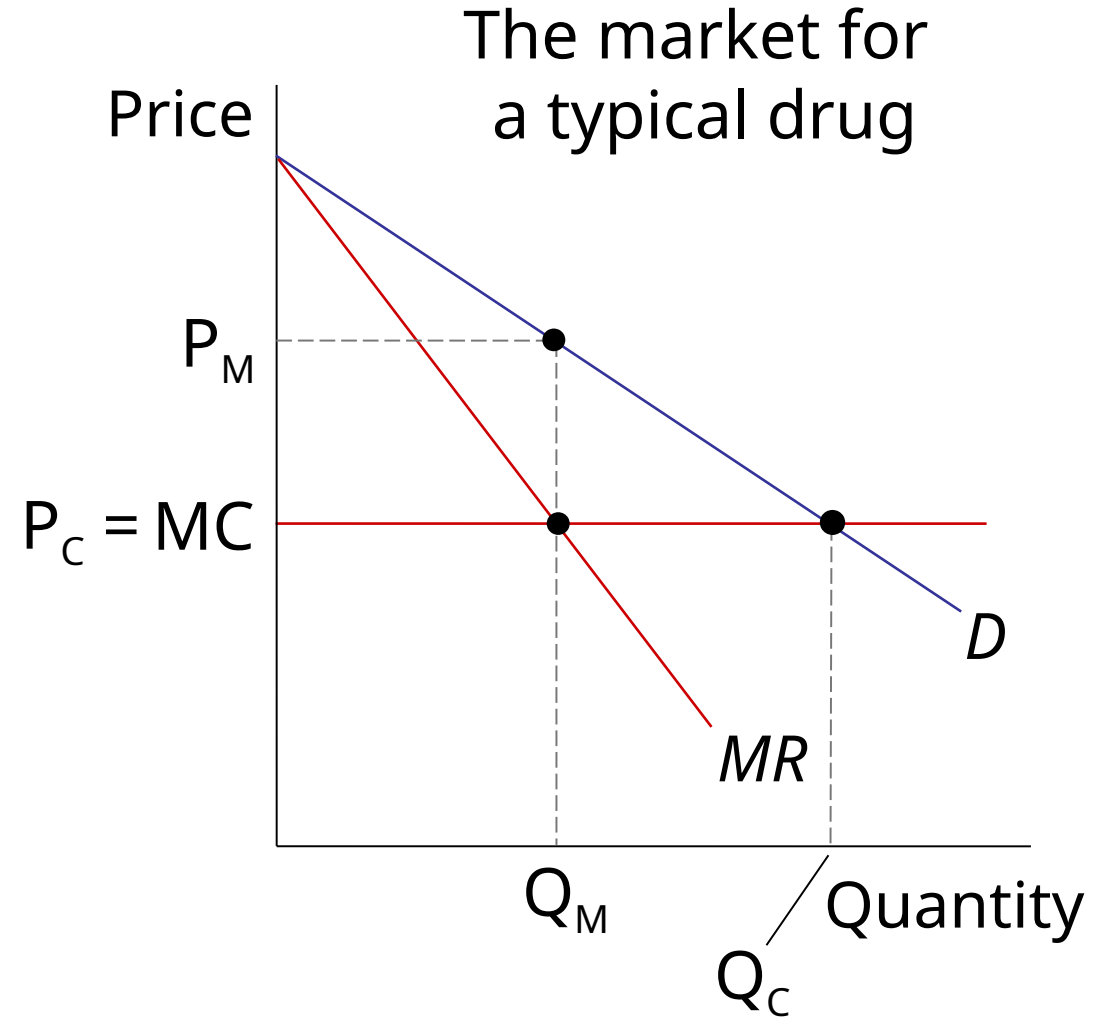
$$P = 16,700 - 2.5(3340)$$

$$\frac{\partial TR}{\partial Q} = 16700 - 5Q = 0$$

$$P = \$8350$$

Monopoly and Generic Drugs

- Patents on new drugs give a temporary monopoly to the seller.
- Notice that the monopoly creates a deadweight loss (a loss in surplus that no one picks up) by charging the monopoly price.
- When the patent expires, the market becomes competitive, generics appear.
- The competitive price sets $MC = P$, and the deadweight loss is erased.



Monopoly Constant Marginal Cost Example (not on a quiz or test)

$$TR = 280Q - 0.005Q^2$$

$$MR = 280 - 0.01Q$$

$$MC = \$100$$

Calculate price, output and profit.

$$MR = MC$$

$$D = AR = P = 280 - 0.005Q$$

$$\pi = TR - TC$$

$$280 - 0.01Q = 100$$

$$P = 280 - 0.005(18000)$$

$$TR = 190 * 18000 = \$3,420,000$$

$$180 = 0.01Q$$

$$P = \$190$$

$$TC = 100 * 18000 = \$1,800,000$$

$$Q = 18000$$

$$\pi = \$1,620,000$$

What price and profit levels would prevail if firms could enter and make the market competitive? (not on a quiz or test)

$$P = MC$$

$$280 - 0.005Q = 100$$

$$180 = 0.005Q$$

$$Q = 36,000$$

$$P = 100$$

$$\pi = 0$$

Constant Marginal Cost Example 2 (not on a quiz or test)

$$P = \$5,000 - 0.05Q$$

$$MR = 5,000 - 0.1Q$$

$$MC = \$4000$$

$$MR = MC$$

$$5000 - 0.1Q = 4000$$

$$Q = 10000$$

$$P = 5000 - 0.05(10000) = \$4,500$$

$$TR = 4000 * 10000 = \$40,000,000$$

$$TC = 4000 * 10000 = \$40,000,000$$

$$\pi = \$5,000,000$$

Calculate the price, output and profit.

Calculate the deadweight loss.

$$P = MC$$

$$5000 - 0.05Q = 4000$$

$$Q = 20,000$$

$$DWL = \frac{1}{2}[(20000 - 10000) * (4500 - 4000)]$$

$$DWL = \$2,500,000$$

Monopoly and Risk

Monopolies are not immune to risk. When expected returns don't match perceived risk, capital moves elsewhere.

- **Compensating Risk Differential:**

- Investors demand higher returns to hold riskier assets.
- This excess return is the **risk premium**:

$$\text{Risk premium} = E[r_{\text{monopoly}}] - r_{\text{safe}}$$

- **Monopoly Stocks May Carry Hidden Risk:**

- Regulatory crackdowns (e.g., antitrust)
- Tech disruption (e.g., AI vs. search engines)
- Public backlash (e.g., big tech scrutiny)
- Overreliance on one product or IP

- **Investor Response:**

- If monopoly stock returns fall or stabilize but risks increase, risk-adjusted returns shrink.
- Rational investors rotate to safer or more dynamic assets (e.g., from big pharma to diversified ETFs).

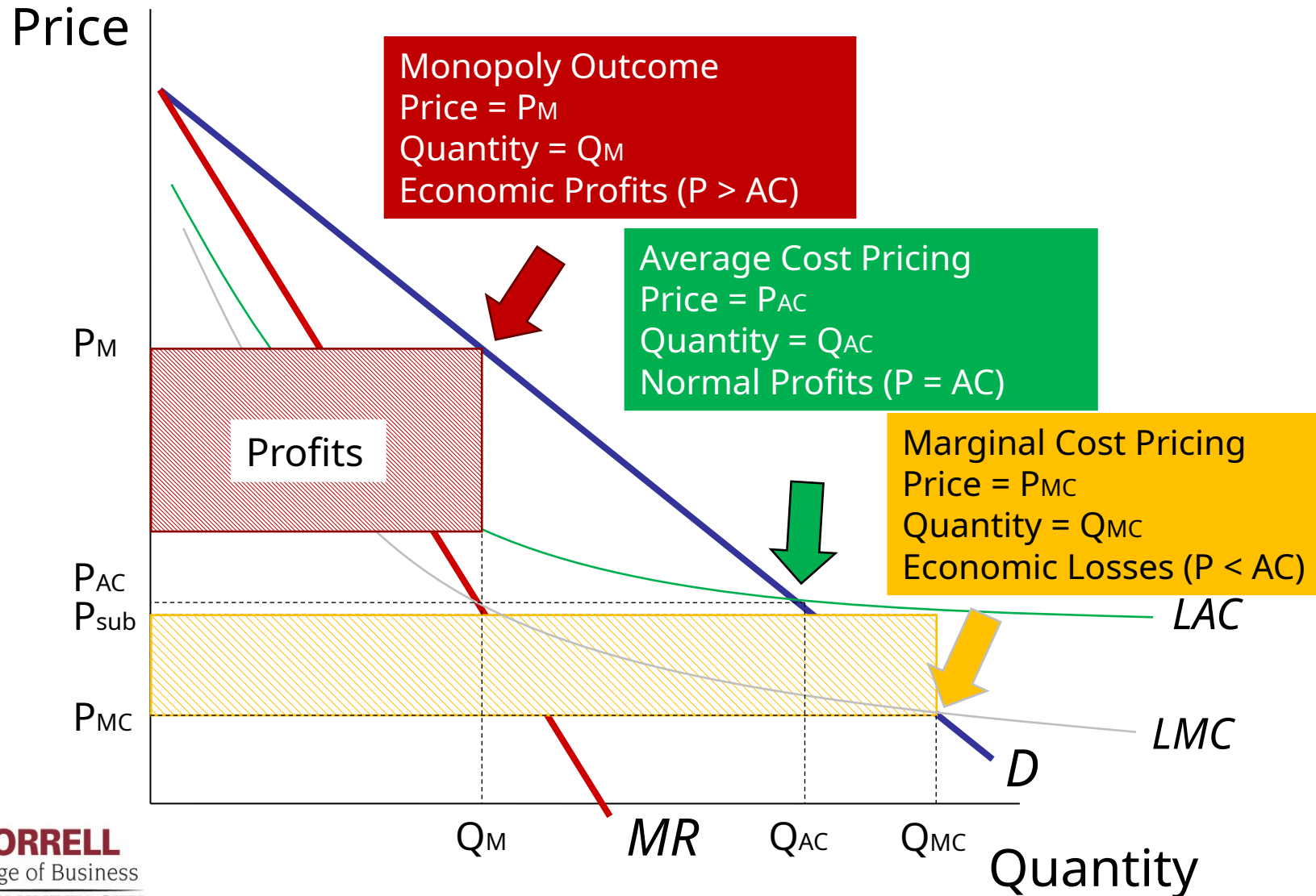
NYC Taxis

- NYC taxicab drivers require a medallion to operate a cab.
- These medallions are in limited number, thus giving rise to economic profits.
- The number of medallions in 1937 was 11,787 and had a discounted value of \$10. By 2009, only 13,257 were available with a discounted value of \$800,000.
- Because NYC issues these medallions it has become difficult to enter into the market, as evidenced by the taxi lobby blocking an increase in the number of medallions.
- Radio cabs (cabs that can only respond to radio calls) have been able to eat away at the monopoly profits of taxis. Growth in the 1980s, eroded 32% of the taxi owners' profits by 1993 with another 11% by 2001.

Monopoly Regulation

- Dilemma of Natural Monopoly
 - Monopoly has the potential for efficiency.
 - Unregulated monopoly can lead to economic profits and underproduction.
- Utility Price and Profit Regulation
 - Regulation is sometimes used to improve monopoly market performance.
 - Substituting bureaucratic decisions for market interactions is risky and costly.

Monopoly Regulation



Case Study: Amtrak

- Amtrak is a publicly funded railroad service for passengers.
- Passenger service for railroads is very costly, usually to the point where it generates losses for the company.
- There is a large upfront fixed cost for the service, with declining average costs.
- In the fiscal year 2018, Amtrak earned approximately \$3.4 billion in revenue and had expenses totaling \$4.8 billion.
- According to [Amtrak](#),
 - “No country in the world operates a passenger rail system without some form of public support for capital costs and/or operating expenses”
- It served 31.7 million passengers during this time.
- Amtrak faces the classic regulatory dilemma.

Case Study: Amtrak

- The losses experienced by Amtrak occur on a yearly basis. The U.S. government has routinely paid more than \$1 billion in subsidies per year.
- From 1972 to 2014, [Amtrak has received more than \\$45 billion in subsidies.](#)
- Is this a good deal?
- According to a [policy option by the CBO](#), eliminating the subsidies will save an estimated \$15 billion between 2014 – 2023.
- One solution was to turn over less profitable routes to the states and let them pick up the subsidy tab.

Competitive Equilibrium vs. Monopoly Erosion

	Competitive Industry	Monopoly / Market Power
Entry Barriers	Low – anyone can enter	High – patents, scale, regulation
Profit in Long Run	Zero economic profit ($P = ATC$)	Eroded over time via imitation and innovation
Mean Reversion?	Strong – profits attract entry, losses push exit	Yes – due to entry pressure, tech shifts, or legal action
Wage/Risk Differentials	Compensate for disamenities (e.g., danger, hours)	Investors demand risk premia; workers may seek safer roles
Investor Behavior	Capital shifts across industries based on returns	Investors exit when risk-adjusted returns shrink
Sustainability of Profits	Unsustainable – competitive pressure eliminates surplus	Temporarily sustainable – eroded unless protected

Key Takeaways

- Perfect competition leads to zero long-run profits due to free entry/exit and market forces.
- Profits in competitive markets exhibit mean reversion — unusually high (or low) returns don't last.
- Compensating wage differentials pay workers more to accept unpleasant or risky jobs.
- Risk premia compensate investors for holding volatile assets like stocks.
- When risk-adjusted returns shrink, rational investors move to safer assets.
- Monopolies earn profit due to barriers to entry, but face erosion through:
 - Direct market entry
 - Product imitation
 - Substitution and tech leapfrogging
- Real-world examples: Pharma after patents, taxis vs. rideshares, Blockbuster vs. Netflix

Knowledge Check

1. In perfect competition, what happens to profits in the long run?
2. What is meant by “mean reversion” of profits?
3. Why do coal miners typically earn more than retail workers?
4. What causes investors to shift out of risky assets like stocks?
5. How do new competitors erode monopoly profits over time?

Knowledge Check Answers

1. They fall to zero (normal rate of return).
2. Profits regress to the average as firms enter/exit.
3. They receive a compensating wage differential for risk and discomfort.
4. Shrinking risk premia — returns no longer justify risk.
5. Through imitation, substitution, tech advances, or market entry.

Consultant's Challenge: Protecting Profits

Scenario:

- You're advising a successful mobile car wash startup that earned big profits last year. Now, three new competitors have entered the space, and your client is starting to feel the squeeze.

Your Task:

1. Collaborate with 1-2 of your classmates to answer these questions.
2. How would you help them understand what's happening through the lens of profit erosion and mean reversion?
3. What advice would you give on how to defend (or justify) above-average returns through differentiation, risk, or barriers to entry?